Can a Cheap Text Monitor Tell Right from Wrong Reasoning?

CSI 436/536 Machine Learning (Fall 2026) — Project P Midterm One-Pager Group: Jiayu Liao, Zeyu Pan, Yuhan Sun Oct 20, 2026

Research question (fixed by the project).

Do pretrained representations improve downstream ML tasks, particularly under limited training data? We turn the yes/no question into a measurement problem. With the class algorithm g and training data fixed, the *only* varied factor is the feature source φ, and we report the gap function

$$\Delta(\theta) \;=\; L_{\text{test}}\big(g \circ \varphi_{\text{simple}};\, \theta\big) \;-\; L_{\text{test}}\big(g \circ \varphi_{\text{pretrained}};\, \theta\big), \qquad \theta = (\text{train size } n,\ \text{shift},\ \text{perturbation},\ \text{input construction}).$$

All reported Δ values are AUROC differences (embedding − TF-IDF), i.e. $L = 1 - \text{AUROC}$ in the formula above: $\Delta > 0$ favors pretrained features.

Task and data.

A cheap *text monitor* for math reasoning: given a problem and one solution step, classify the step as correct or incorrect. **PRM800K** (OpenAI, MIT): human-labeled step-level correctness on MATH solutions — 911,867 labeled steps (27.9% incorrect) after our data preparation; grouped by problem into 632,422 train / 88,872 val (reserved, unused this round) / 190,573 test steps. **Math-Shepherd** (GSM8K, automatically labeled, 613,666 steps): transfer test set only. Rule-based perturbed variants (typos, number formatting, LaTeX style): robustness test sets. Labels come with the datasets; no LLM labeling is used.

Method (strictly within the rules).

Pretrained embeddings — `all-MiniLM-L6-v2`, `bge-base`, `bge-large` — are *frozen* feature extractors only (forward pass, no fine-tuning). On top we train only models covered in class: logistic regression (L7) and linear SVM (L11). Control arm: TF-IDF (word / char n-grams) with the same classifiers. Protocol: input = problem + current step (main setting; input construction ablated in E7); **group split by problem** (no step of one problem crosses train/test); primary metrics **balanced accuracy / AUROC** (labels are imbalanced); 10 stratified re-samples per n, mean ± CI; C selected by grouped k-fold CV inside the training set; test set touched once; all seeds fixed.

Current progress (three axes complete + reviewer-driven input ablation).

$\Delta(n)$**, sample efficiency** (problem+step input): at n=20 all features sit near chance (≈ 0.51 AUROC); from n=50 TF-IDF pulls ahead monotonically; at full data $\Delta = -0.045$ to -0.064 (SVM, CI excludes 0) — no in-domain benefit *in this input construction.* **E7 input ablation** (reviewer question: features, or input?): truncation is real but tiny (median input 89 tokens; MiniLM truncates 1.6% of rows, bge 0.04%). Dropping the problem text (step-only) lifts embeddings by +0.030–+0.061 vs. only +0.012–+0.016 for TF-IDF — *the problem text hurts embeddings far more than TF-IDF*, consistent with pooling dilution (not yet isolated as the sole mechanism). Step-only bge-large (0.711) draws level with TF-IDF (0.708; point estimate from the full-data run, CI to follow). *The in-domain negative* Δ *is mostly an input-construction effect, not a feature property.* Δ**(shift), generalization**: with step-only input the transfer gain triples — bge-large $\Delta = +0.069$, CI $[0.067, 0.071]$; bge-base +0.049; MiniLM +0.030 (all significant, paired bootstrap; original construction gave bge-large +0.020). Removing the problem text *hurts* TF-IDF transfer (0.551 → 0.527); embeddings also overfit far less (train–test gap 0.040–0.065 vs. 0.153–0.168). Δ**(noise), robustness**: under rule-based perturbations all degradations are small (≤ 0.007 AUROC), embeddings no worse than TF-IDF. Headline: *the value of pretrained features is conditional on input construction — and largest under distribution shift.* Robustness checks: 50/50 re-sampling and `class_weight=balanced` change AUROC by < 0.01.

Challenges and plans.

We caught our own label-construction bug early (chosen-only completions, 99.8% positives, degenerate metrics); everything above is the corrected re-run. The MATH→GSM8K shift is a *compound* shift (problem type, source, labeling method) — reported without single-factor attribution; a decomposition is planned. Cheapest mechanism check next: encode problem and step *separately* and concatenate — keeps context, avoids pooling dilution. Also next: a step-only $\Delta(n)$ curve and the step-only in-domain CI (cached features make both cheap), single-factor shift decomposition, and a final measurement report (effect sizes + three boundary curves, both input constructions).

AI use disclosure (per syllabus §4.8).

The implementation code was written by AI assistants under our direction (also used for language polishing). The research framing, experimental design, data checking, and ablation design were carried out by the group; we have read all of the code and can explain every part of it.

Acknowledgments.

This work uses public datasets and open model weights: the **PRM800K** dataset (OpenAI, MIT license), **Math-Shepherd** (GSM8K), and the frozen sentence-transformers checkpoints `all-MiniLM-L6-v2`, `bge-base-en-v1.5` and `bge-large-en-v1.5` (BAAI). We thank the authors for making them publicly available.